

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Weeks 26-28
6 July - 26 July 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



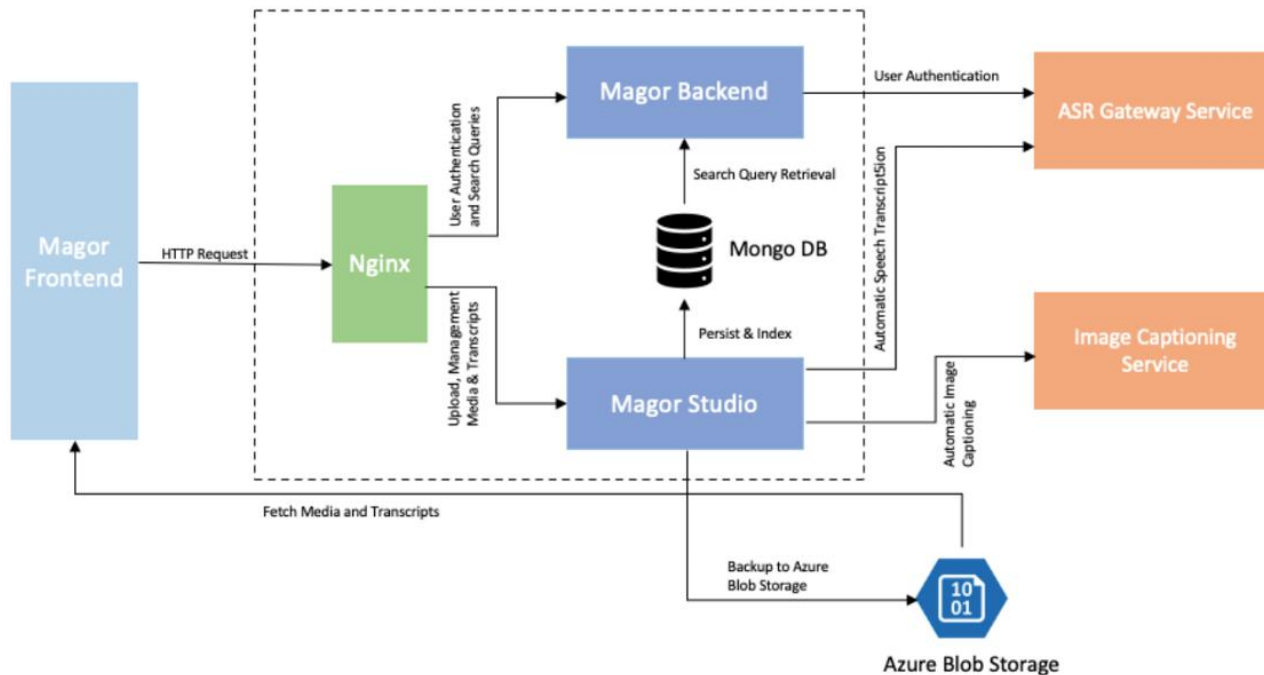
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

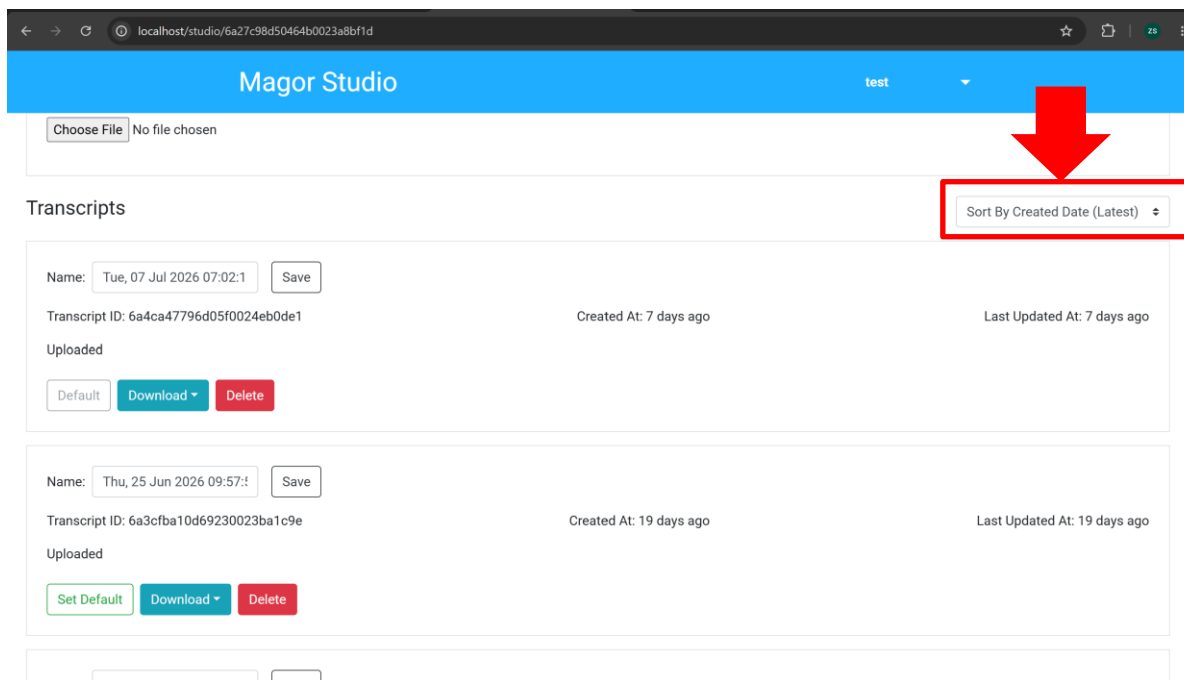
Work Done Over The Weeks:

- Updated Transcript Studio default sorting to display the latest recordings first
- Implemented the "Ask AI" summarization feature
 - **Weeks 26-27:** Implemented authentication token generation and the core summarization workflow
 - **Weeks 27-28:** Enhanced the UI, implemented manual polling, and added database persistence for summary results



Task 1 (Transcript Studio Default Sort by Latest)

- Studio Transcript List:



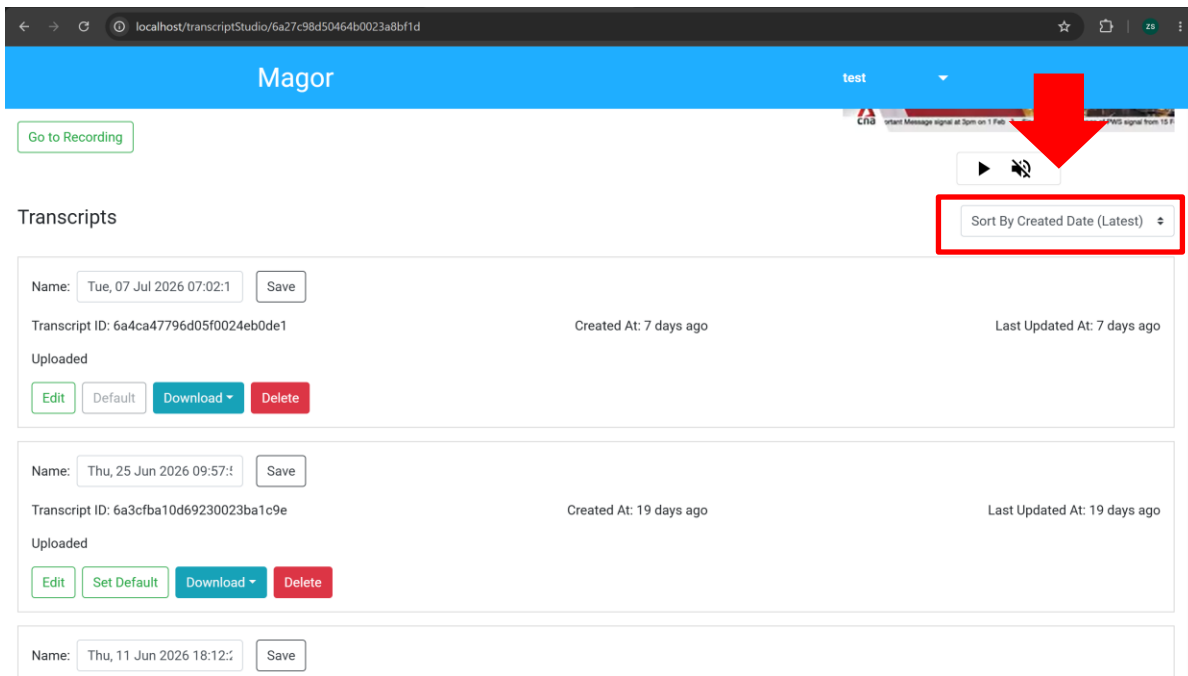
Task 1 (Transcript Studio Default Sort by Latest)

- Transcript Studio Transcript List:

The screenshot shows the Transcript Studio interface. At the top, there's a blue header with the name 'Magor' and a dropdown menu set to 'test'. Below the header, the main content area displays a transcript entry. The entry has a title 'Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4' and a description 'No Description'. A green button labeled 'Go to Recording' is visible. To the right of the text is a video player showing a news broadcast from CNA Asia Tonight. Below the video player, a red arrow points to a dropdown menu labeled 'Sort By Created Date (Earliest)'. Below this, the 'Transcripts' section shows a list of transcripts. The first transcript has a name 'Thu, 11 Jun 2026 18:12:...', a transcript ID '6a2afa8ddbd343002313a179', and was created 'a month ago'. It has buttons for 'Edit', 'Set Default', 'Download', and 'Delete'. The second transcript is partially visible with the name 'Thu, 25 Jun 2026 09:57:...'.

Task 1 (Transcript Studio Default Sort by Latest)

- Transcript Studio Transcript List Fixed:



Task 2 (“Ask AI” Generate Summary Functionality)

- “Ask AI” button:

The screenshot displays the Magor application interface. At the top, there is a blue header bar with the word "Magor" on the left, a "Search" input field in the center, and a "This recording" dropdown menu on the right. Below the header, the main content area is divided into two sections. The left section features a large red graphic with the text "ASIA Tonight" and a "cna" logo. The right section contains a list of topics: "more than A (144p", "market point indonesia fiscal", "ss deepening month", and "for Indonesian shares, the".

Overlaid on the interface is a white dialog box titled "AI Assistant". Inside the dialog, there is a "Summary" section. A red box highlights the text "Implement Actual Summary From SpeechLabs LLM API" with a red arrow pointing down to the "Summary" section. The "Summary" section contains the following text:

Summary
This is a placeholder summary of the transcript. The transcript contains approximately 528 words.

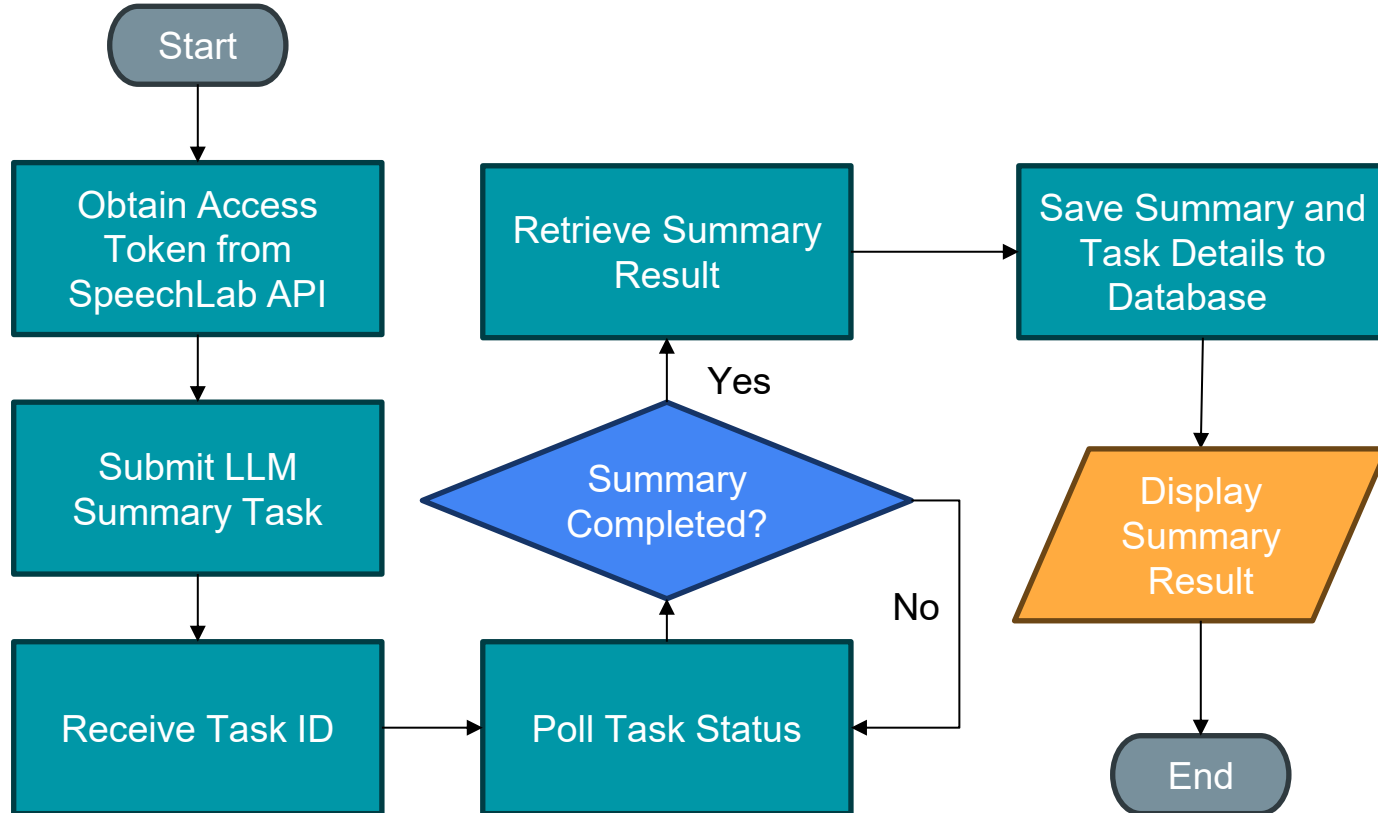
Key Points with Timestamps:

- [1:30] oil and business sharp declines for Indonesian sha...
- [6:52:30] currency has been sliding since August 2025 on con...
- [12:19:10] Uh, Shawnie concerns, confidence, both about centr...

To enable actual AI summarization with key points and timestamps, please configure an AI service (e.g., OpenAI, Azure OpenAI) in the controller implementation. The AI will be instructed to confine queries to the contents of the transcript only.

Task 2 (“Ask AI” Generate Summary Functionality)

- Process Flow:



Task 2 (“Ask AI” Generate Summary Functionality)

- Core summary generation functionality demo:

The screenshot displays the Magor application interface. At the top is a blue header with the 'Magor' logo, a search bar, and a 'This recording' dropdown menu. Below the header, the main content area is split. On the left is a video player showing a news broadcast from CNA Asia Business Tonight, with a cityscape background. The video player includes a progress bar and a volume icon. On the right is a text area containing an AI-generated summary of the video content. The summary title is 'Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4'. Below the title are 'Top Words' including indonesia, concerns, msci, bank, market, point, indonesia, fiscal, president, shares, warning, downgrade, progress, deepening, and month. There are buttons for 'Transcript', 'Show options', and 'Ask AI'. The transcript shows two speakers discussing the decline of Indonesian shares and the impact of MSCI's warning.

Magor Search This recording test

Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4 Edit

Top Words: indonesia concerns msci bank market point indonesia fiscal president shares warning downgrade progress deepening month

Transcript Show options Ask AI

speaker 1 oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30-minute circuit breaker. The Jakarta composite ending down more than 7% after MSCI warned about the market's instability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today, it has been under sustained pressure for months, tumbling to an all-time low against the US dollar.

speaker 1 currency has been sliding since August 2025 on concerns about fiscal discipline or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its

Task 2 (“Ask AI” Generate Summary Functionality)

- Enhanced Summary Generation UI
 - Implement UI handling for different summary generation states
 - Add manual polling support for users to check summary status on demand
 - Handle timeout scenarios and display appropriate user notifications
 - Implement error handling for failed summary generation requests.



Task 2 (“Ask AI” Generate Summary Functionality)

- Enhanced Summary Generation UI

The screenshot displays the Magor AI interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, and a 'This recording' dropdown menu. Below the header, the video player shows a man and a woman in a studio setting. To the right of the video, the title 'Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4' is displayed, along with an 'Edit' button. Below the title, 'Top Words' are listed: party, parliament, opposition, leader, workers, pritam, singh, mister, en, time, position, people, office, seats, budget. A 'Transcript' button is visible, followed by 'Show options' and 'Ask AI'. The transcript itself is divided into segments: 'channel1_S1' (pink background) and 'channel1_S2' (blue background). The video player at the bottom shows '0 matches' and 'Nothing is playing right now.' with a progress bar at 0:00 / 30:42. A speaker icon is located on the right side of the interface.

Magor Search This recording test

Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 Edit

Top Words: party parliament opposition leader workers pritam singh mister en time position people office seats budget

Transcript Show options Ask AI

channel1_S1 so everyone welcome back to another episode of deep dive with me steven and michael tiffany now um this week we're talking about uh the political scene in singapore because the workers party leader pritam singh is no longer singapore's leader of the opposition

channel1_S2 because ah prime minister lawrence wong he has removed mister singh from this role and invited the workers' party to nominate another person to take the seat

channel1_S1 right yeah yeah that's right and this is all coming on the back of mister singh's court convictions for lying to the committee of privileges although whether he had instructed former m p ratio khan to come clean about a lie she told in parliament

0 matches < > Nothing is playing right now. 0:00 / 30:42

Plans for Next Week

- “Ask AI” Button UI adjustments
 - Refine UI with a side-by-side view layout alongside the recording (Improve Usability)
 - Add a custom summary option to support user-defined prompts.
 - Update the default summary prompt to include timestamps and align with YouTube’s “Ask AI” summary format.
- Project Maintenance
 - Review and upgrade Node.js versions along with dependencies

